

8_Matplot&Seaborn_DataRepresentation

February 11, 2025

```
[1]: pip install seaborn
```

```
Collecting seaborn
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in ./lib/python3.13/site-packages (from seaborn) (2.2.1)
Requirement already satisfied: pandas>=1.2 in ./lib/python3.13/site-packages (from seaborn) (2.2.3)
Collecting matplotlib!=3.6.1,>=3.4 (from seaborn)
  Downloading matplotlib-3.10.0-cp313-cp313-macosx_11_0_arm64.whl.metadata (11 kB)
Collecting contourpy>=1.0.1 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading contourpy-1.3.1-cp313-cp313-macosx_11_0_arm64.whl.metadata (5.4 kB)
Collecting cycler>=0.10 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading fonttools-4.55.3-cp313-cp313-macosx_10_13_universal2.whl.metadata (165 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading kiwisolver-1.4.8-cp313-cp313-macosx_11_0_arm64.whl.metadata (6.2 kB)
Requirement already satisfied: packaging>=20.0 in ./lib/python3.13/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (24.2)
Collecting pillow>=8 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading pillow-11.1.0-cp313-cp313-macosx_11_0_arm64.whl.metadata (9.1 kB)
Collecting pyparsing>=2.3.1 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading pyparsing-3.2.1-py3-none-any.whl.metadata (5.0 kB)
Requirement already satisfied: python-dateutil>=2.7 in ./lib/python3.13/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in ./lib/python3.13/site-packages (from pandas>=1.2->seaborn) (2024.2)
Requirement already satisfied: tzdata>=2022.7 in ./lib/python3.13/site-packages (from pandas>=1.2->seaborn) (2024.2)
Requirement already satisfied: six>=1.5 in ./lib/python3.13/site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
Downloading seaborn-0.13.2-py3-none-any.whl (294 kB)
Downloading matplotlib-3.10.0-cp313-cp313-macosx_11_0_arm64.whl (8.0 MB)
```

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8.0/8.0 MB
1.2 MB/s eta 0:00:0000:0100:01
Downloading contourpy-1.3.1-cp313-cp313-macosx_11_0_arm64.whl (255 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.55.3-cp313-cp313-macosx_10_13_universal2.whl (2.8 MB)
2.8/2.8 MB
1.2 MB/s eta 0:00:00a 0:00:01
Downloading kiwisolver-1.4.8-cp313-cp313-macosx_11_0_arm64.whl (65 kB)
Downloading pillow-11.1.0-cp313-cp313-macosx_11_0_arm64.whl (3.1 MB)
3.1/3.1 MB
1.1 MB/s eta 0:00:00a 0:00:01
Downloading pyparsing-3.2.1-py3-none-any.whl (107 kB)
Installing collected packages: pyparsing, pillow, kiwisolver, fonttools, cycler,
contourpy, matplotlib, seaborn
Successfully installed contourpy-1.3.1 cycler-0.12.1 fonttools-4.55.3
kiwisolver-1.4.8 matplotlib-3.10.0 pillow-11.1.0 pyparsing-3.2.1 seaborn-0.13.2
Note: you may need to restart the kernel to use updated packages.

```

```
[2]: import numpy as np
import pandas as pd
```

```
[3]: df_sales = pd.read_excel('linechart.xlsx')
df_sales.head()
```

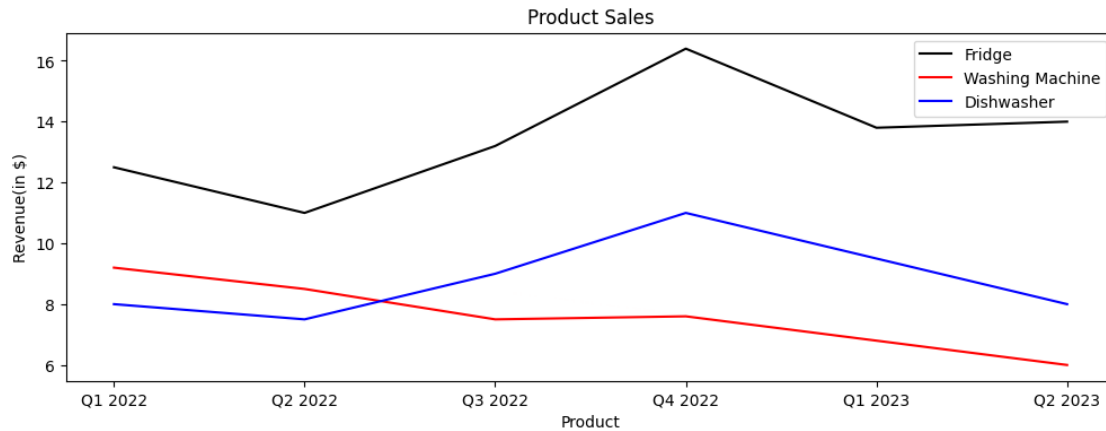
```
[3]:
```

	Quarter	Fridge	Dishwasher	Washing Machine
0	Q1 2022	12.5	8.0	9.2
1	Q2 2022	11.0	7.5	8.5
2	Q3 2022	13.2	9.0	7.5
3	Q4 2022	16.4	11.0	7.6
4	Q1 2023	13.8	9.5	6.8

```
[59]: from matplotlib import pyplot as plt
plt.figure(figsize = (12,4))
plt.plot(df_sales['Quarter'],df_sales['Fridge'], color = 'black', label = 'Fridge')
plt.plot(df_sales['Quarter'],df_sales['Washing Machine'],color = 'red', label = 'Washing Machine')
plt.plot(df_sales['Quarter'],df_sales['Dishwasher'], color = 'blue', label = 'Dishwasher')

plt.title('Product Sales')
plt.xlabel('Product')
plt.ylabel('Revenue(in $)')

plt.legend()#this shows the top right labels
plt.show()
```

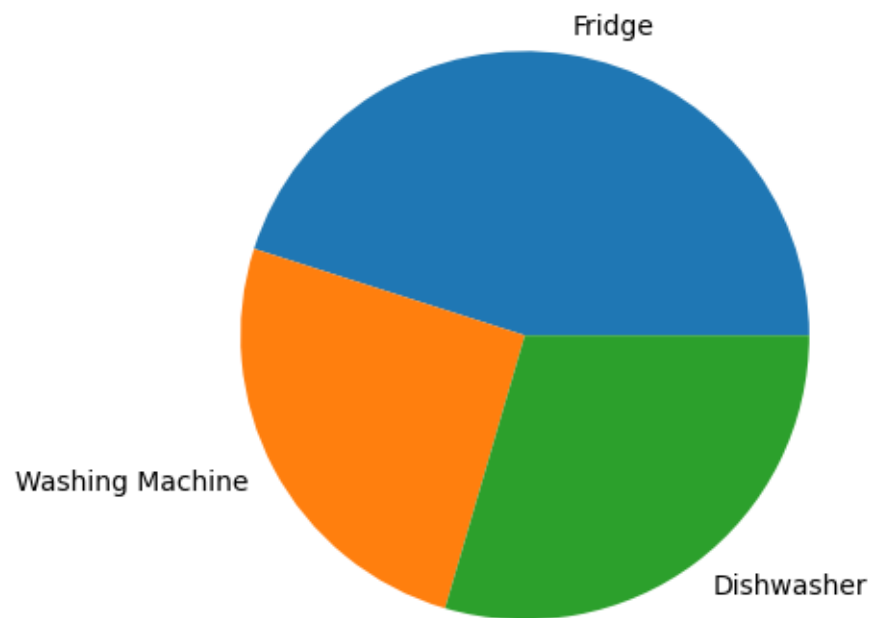


```
[32]: total_sales = df_sales[['Fridge', 'Washing Machine', 'Dishwasher']].sum()
      total_sales.index
```

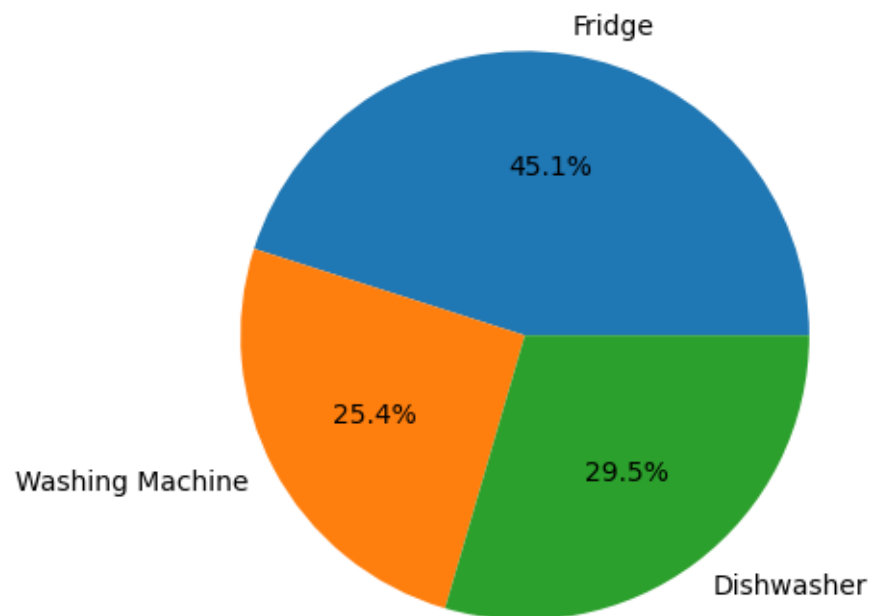
```
[32]: Index(['Fridge', 'Washing Machine', 'Dishwasher'], dtype='object')
```

```
[35]: plt.pie(total_sales, labels = total_sales.index)
```

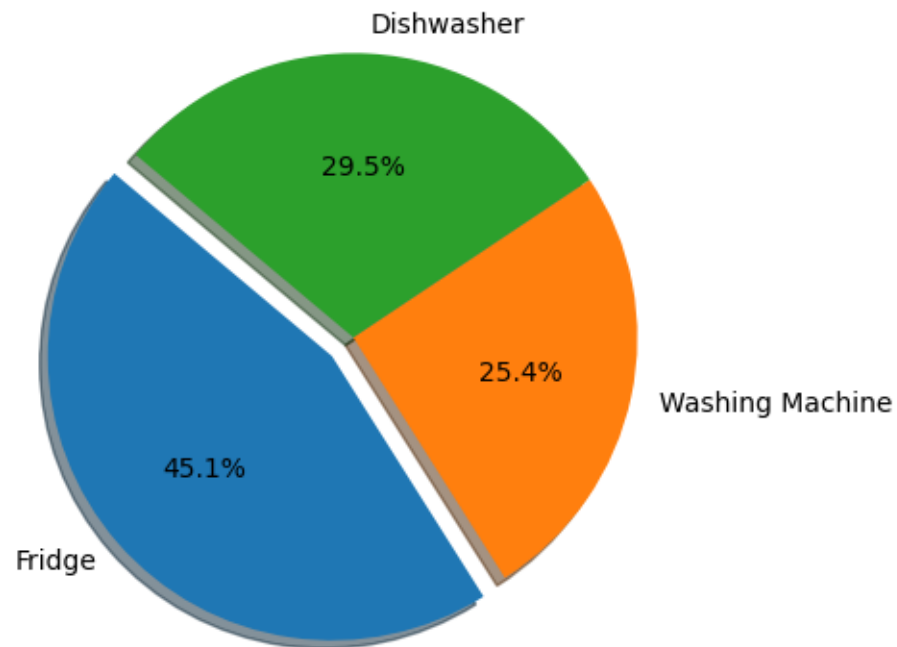
```
[35]: ([<matplotlib.patches.Wedge at 0x130f1d310>,
      <matplotlib.patches.Wedge at 0x130f1d590>,
      <matplotlib.patches.Wedge at 0x130f1d810>],
      [Text(0.1697005203681718, 1.0868310509857417, 'Fridge'),
      Text(-0.9714431235446588, -0.5160409457763957, 'Washing Machine'),
      Text(0.6597307378801044, -0.8802018822384857, 'Dishwasher')])
```



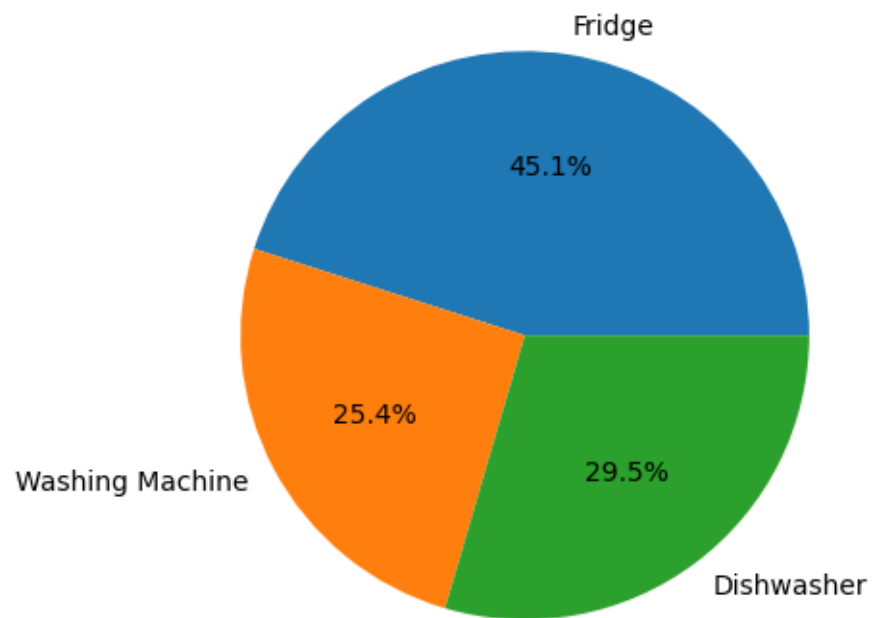
```
[40]: plt.pie(total_sales, labels = total_sales.index, autopct = '%1.1f%%')  
plt.show() #this hides the descriptions thats seen before pie chart
```



```
[43]: plt.pie(total_sales, labels = total_sales.index, autopct = '%1.1f%%', explode =  
      ↪(0.1,0,0), shadow = True,startangle = 140)  
plt.show() #this hides the descriptions thats seen before pie chart
```

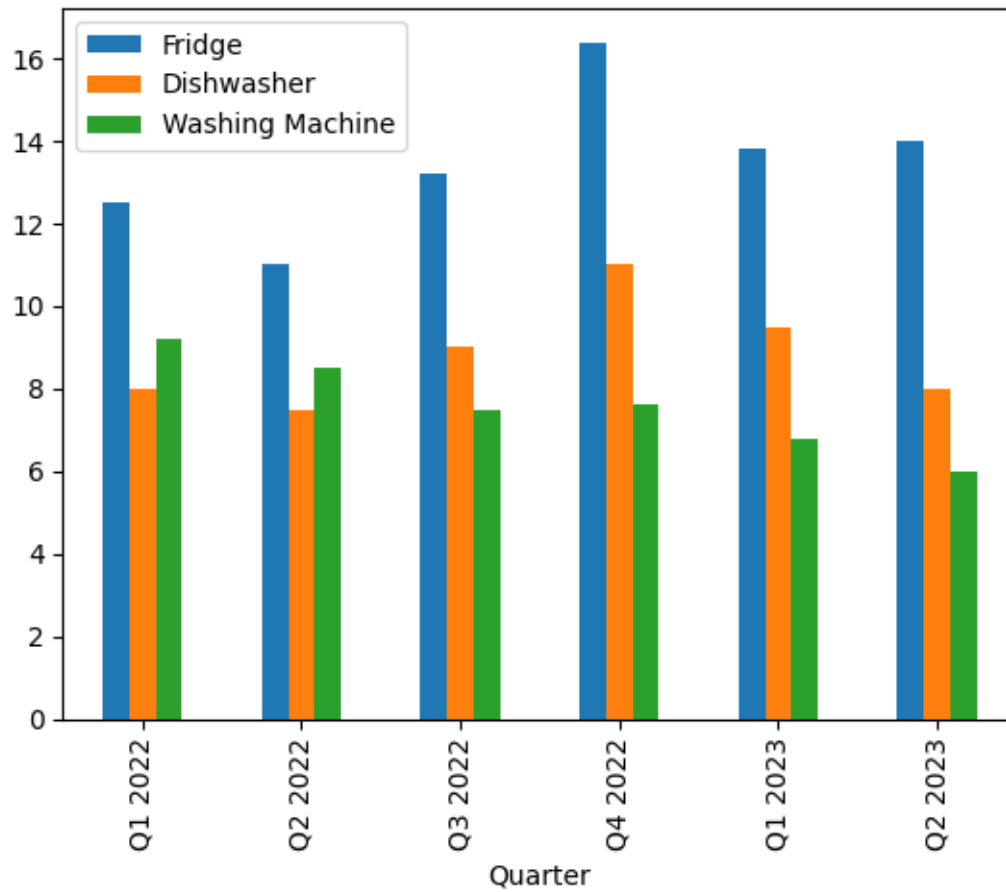


```
[63]: plt.pie(total_sales, labels = total_sales.index, autopct = '%1.1f%%')  
plt.show()
```

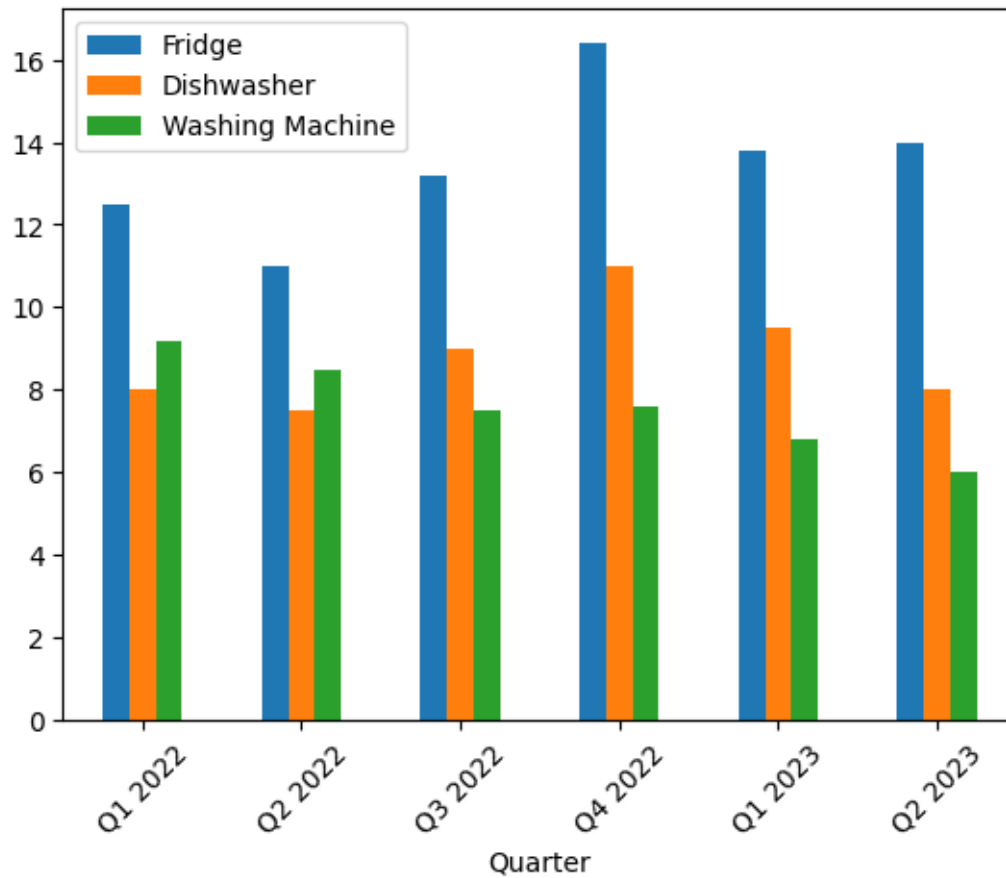


```
[64]: df_sales.plot(kind = 'bar', x = 'Quarter')
```

```
[64]: <Axes: xlabel='Quarter'>
```



```
[68]: df_sales.plot(kind = 'bar', x = 'Quarter')  
plt.xticks(rotation= 45)  
plt.show()
```



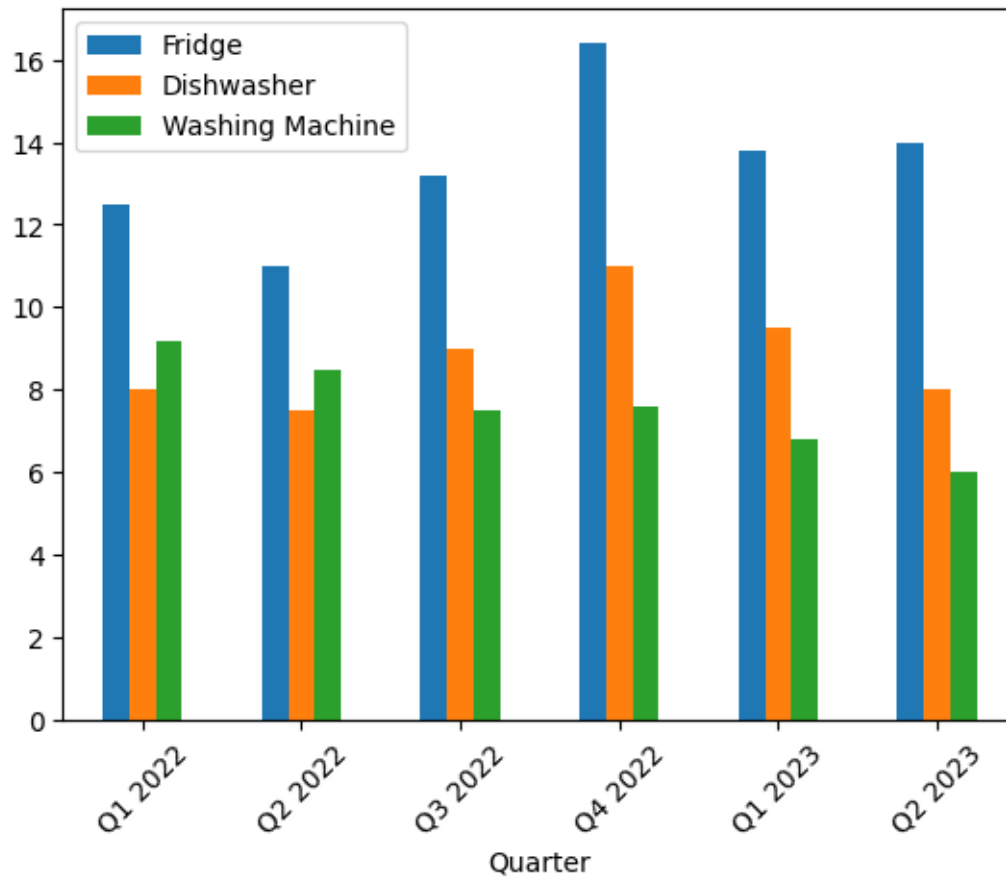
0.1 If we change the index it automatically uses the index as x-axis

```
[70]: df_sales_2 = df_sales.set_index('Quarter')
df_sales_2
```

```
[70]:
```

	Fridge	Dishwasher	Washing Machine
Quarter			
Q1 2022	12.5	8.0	9.2
Q2 2022	11.0	7.5	8.5
Q3 2022	13.2	9.0	7.5
Q4 2022	16.4	11.0	7.6
Q1 2023	13.8	9.5	6.8
Q2 2023	14.0	8.0	6.0

```
[74]: df_sales_2.plot(kind = 'bar')
plt.xticks(rotation = 45)
plt.show()
```



```
[75]: import seaborn as sns
```

```
[89]: df_score = pd.read_excel('histograms.xlsx')
df_score['Exam_Score'].value_counts()
```

```
[89]: Exam_Score
```

```
59    2
79    2
45    1
50    1
95    1
80    1
75    1
72    1
49    1
42    1
91    1
66    1
55    1
```

```

62    1
77    1
76    1
71    1
67    1
48    1
23    1
84    1
90    1
53    1
41    1
93    1
0     1
92    1
9     1
Name: count, dtype: int64

```

```

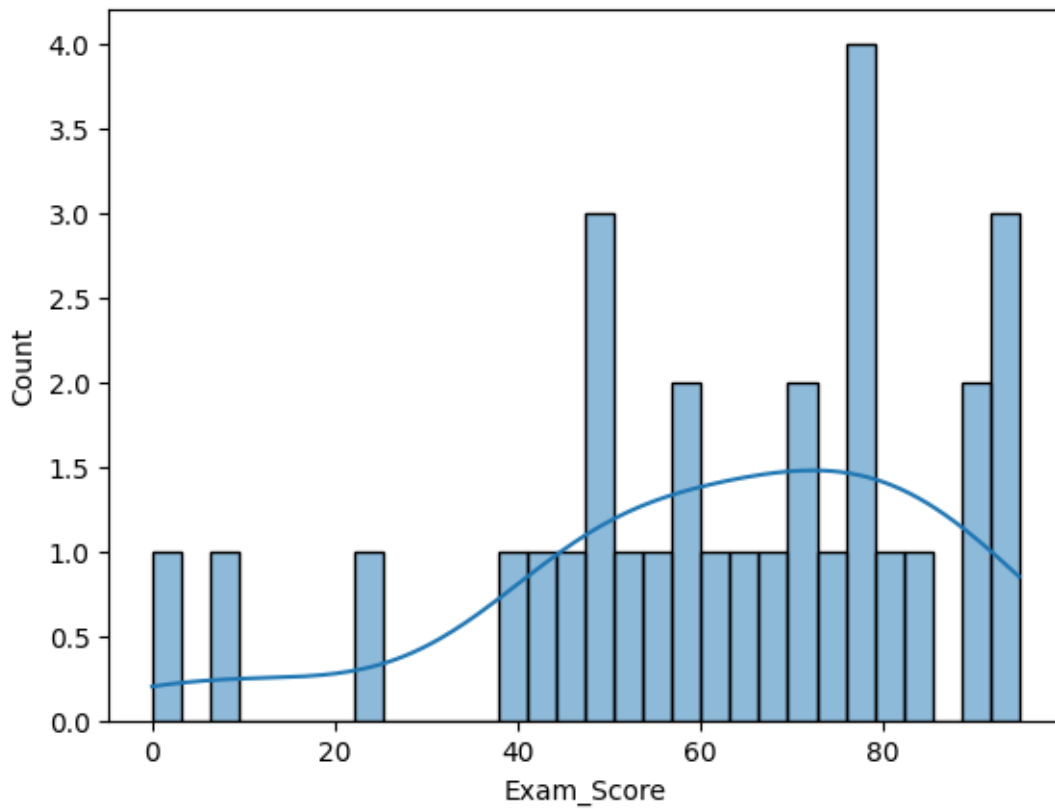
[90]: sns.histplot(df_score['Exam_Score'],bins = 30, kde = True)
      #kde = kernel density estimation

```

```

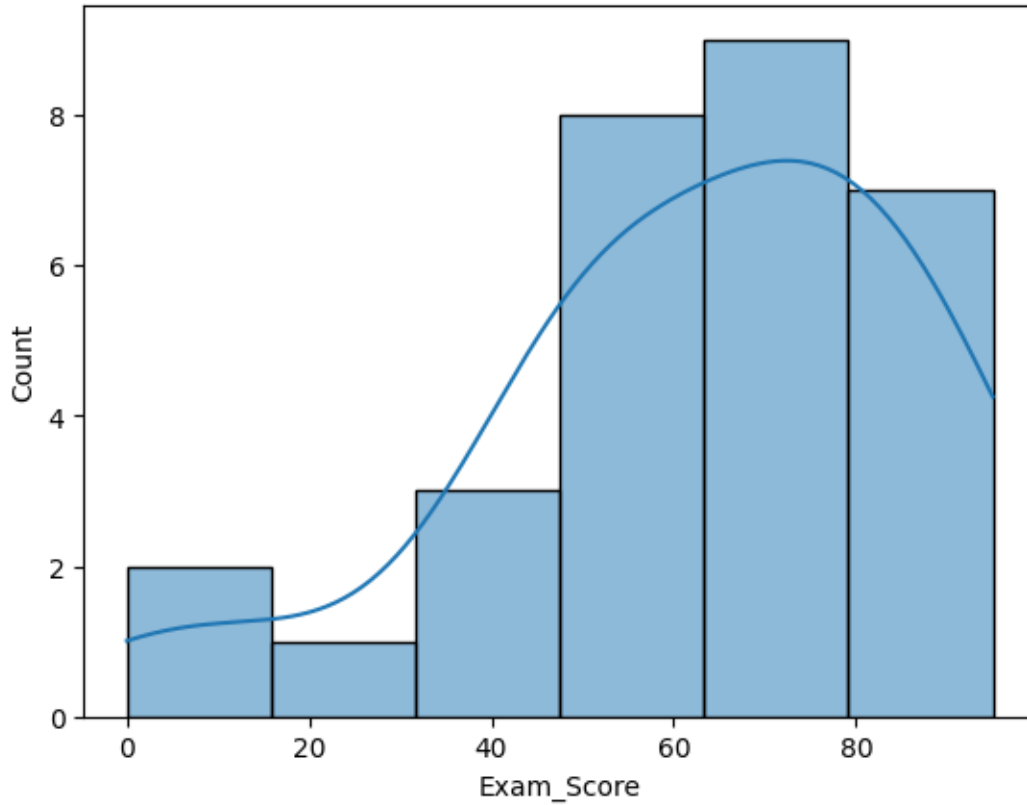
[90]: <Axes: xlabel='Exam_Score', ylabel='Count'>

```



```
[79]: sns.histplot(data = df_score, x = 'Exam_Score',kde = True)
      #kde = kernel density estimation
```

```
[79]: <Axes: xlabel='Exam_Score', ylabel='Count'>
```

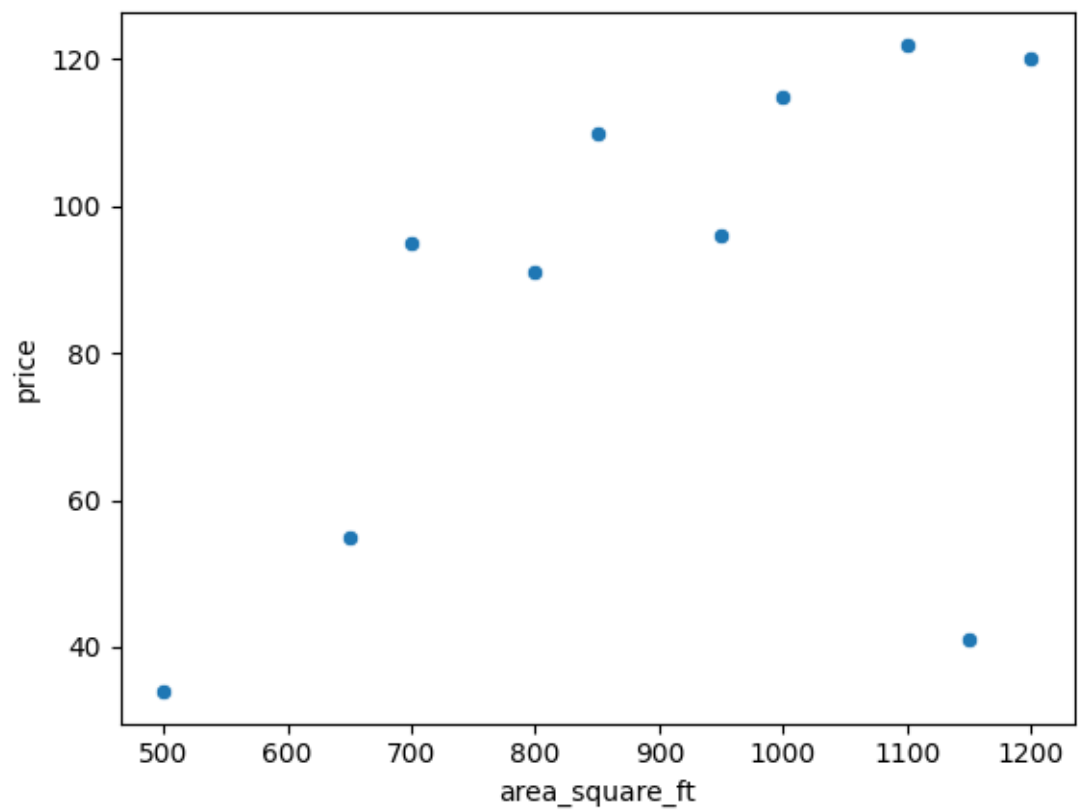


```
[82]: df = pd.read_excel('scatter_plot.xlsx')
      df.head()
```

```
[82]:   area_square_ft  price
0         650      55
1         800      91
2         950      96
3        1100     122
4         500      34
```

```
[92]: sns.scatterplot(data = df, x = 'area_square_ft', y = 'price')
```

```
[92]: <Axes: xlabel='area_square_ft', ylabel='price'>
```



[]: